

Amazon SQS Dead Letter Queue Lab

To Begin with the Lab

Summary of the Lab

In this lab, an existing SQS queue is configured with a Dead Letter Queue (DLQ). A Lambda function is already attached to the main queue and is intentionally modified to throw an exception whenever it receives a message.

When a message is sent to the main queue, Lambda tries to process it but fails. SQS retries the message based on the configured **Maximum Receives** value. Once all retry attempts are exhausted, SQS automatically moves the failed message to the Dead Letter Queue, where it can later be inspected or reprocessed.

Understanding Dead Letter Queue (DLQ)

A **Dead Letter Queue (DLQ)** is a special SQS queue used to store messages that cannot be processed successfully after multiple retry attempts.

Instead of losing failed messages or retrying forever, SQS safely stores them inside the DLQ so that developers or administrators can investigate the failure.

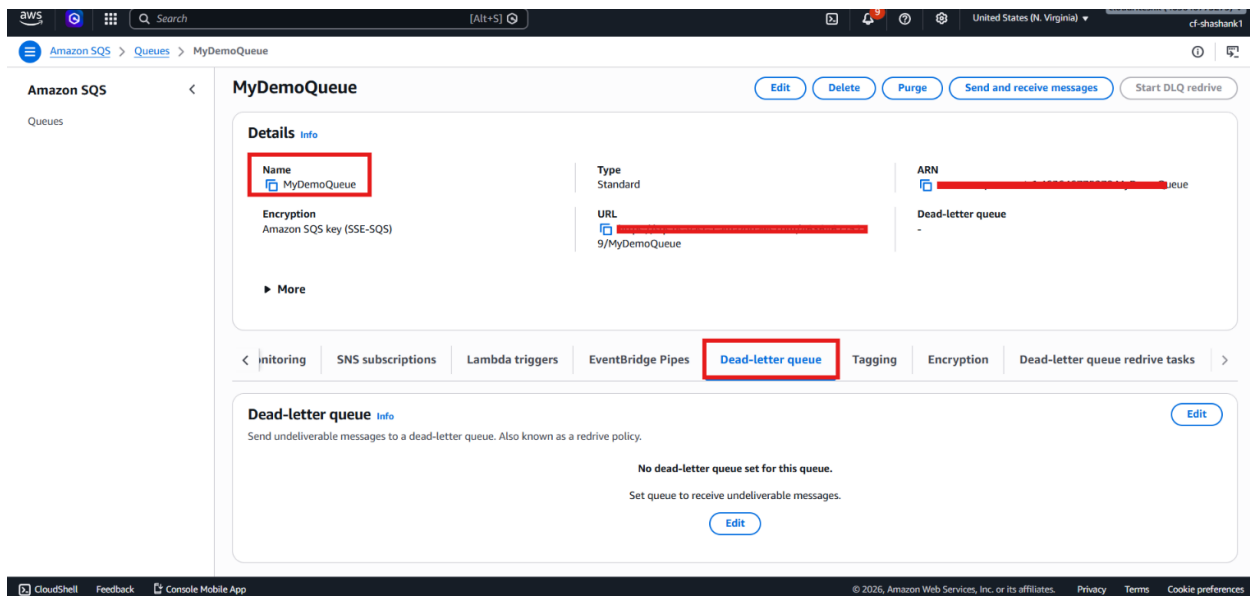
Why use a DLQ?

- Prevents infinite retry loops.
- Stores failed messages safely.
- Makes debugging easier.
- Allows messages to be reprocessed later after fixing the issue.

Lab Steps

Step 1 – Open the Existing SQS Queue

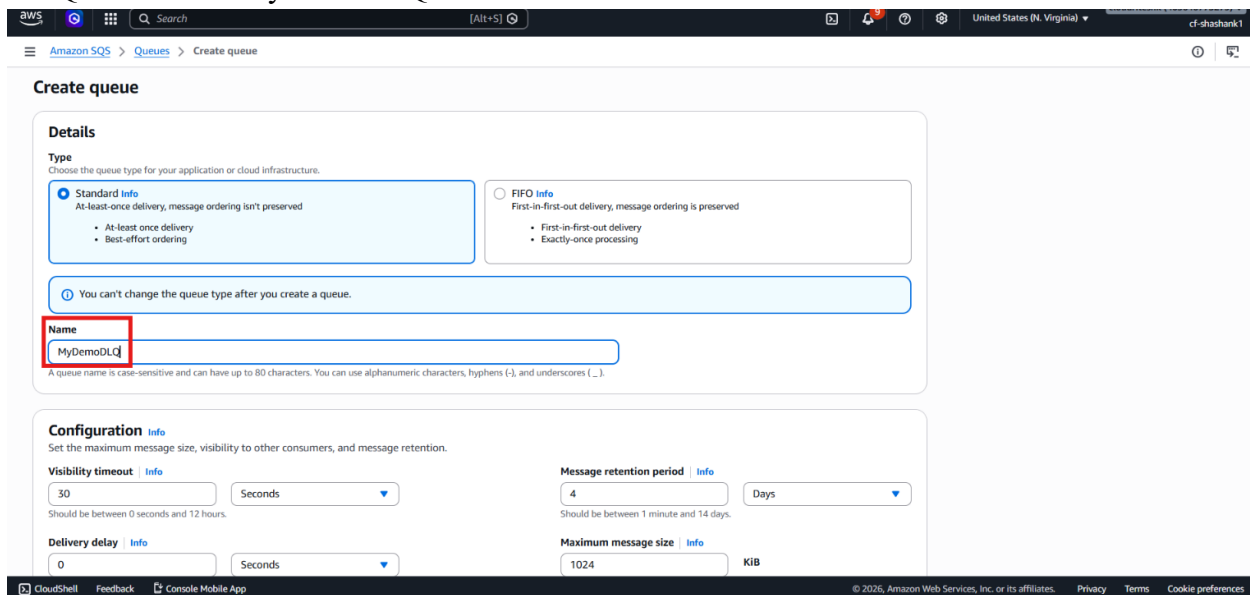
- Navigate to Amazon SQS
- Open the existing **Main Queue**: MyDemoQueue.
- Select the **Dead Letter Queue** tab.



- Initially, no DLQ is configured for the queue.

Step 2 – Create a Dead Letter Queue

- Create a new queue.
- Queue Type: Standard
- Queue Name: MyDemoDLQ



- Leave all remaining settings as default.
- Click Create Queue

Redrive allow policy - Optional [Info](#)
Identify which source queues can use this queue as the dead-letter queue.
Select which source queues can use this queue as the dead-letter queue.
☒ Disabled
☐ Enabled

Dead-letter queue - Optional [Info](#)
Send undeliverable messages to a dead-letter queue.
Set this queue to receive undeliverable messages.
☒ Disabled
☐ Enabled

Tags - Optional [Info](#)
A tag is a label assigned to an AWS resource. Use tags to search and filter your resources or track your AWS costs.
Key Value - optional [Remove](#)
[Add new tag](#)
You can add 49 more tags.

[Cancel](#) [Create queue](#)

- The new queue will act as the Dead Letter Queue.

Step 3 – Attach the DLQ to the Main Queue

- Return to the **Main Queue**.
- Click on **Edit**.

Amazon SQS > **Queues** > **MyDemoQueue**

[Edit](#) [Delete](#) [Purge](#) [Send and receive messages](#) [Start DLQ redrive](#)

Details [Info](#)

Name MyDemoQueue	Type Standard	ARN arn:aws:sqs:us-east-1:123456789012:MyDemoQueue
Encryption Amazon SQS key (SSE-SQS)	URL https://sqs.us-east-1.amazonaws.com/123456789012/MyDemoQueue	Dead-letter queue -

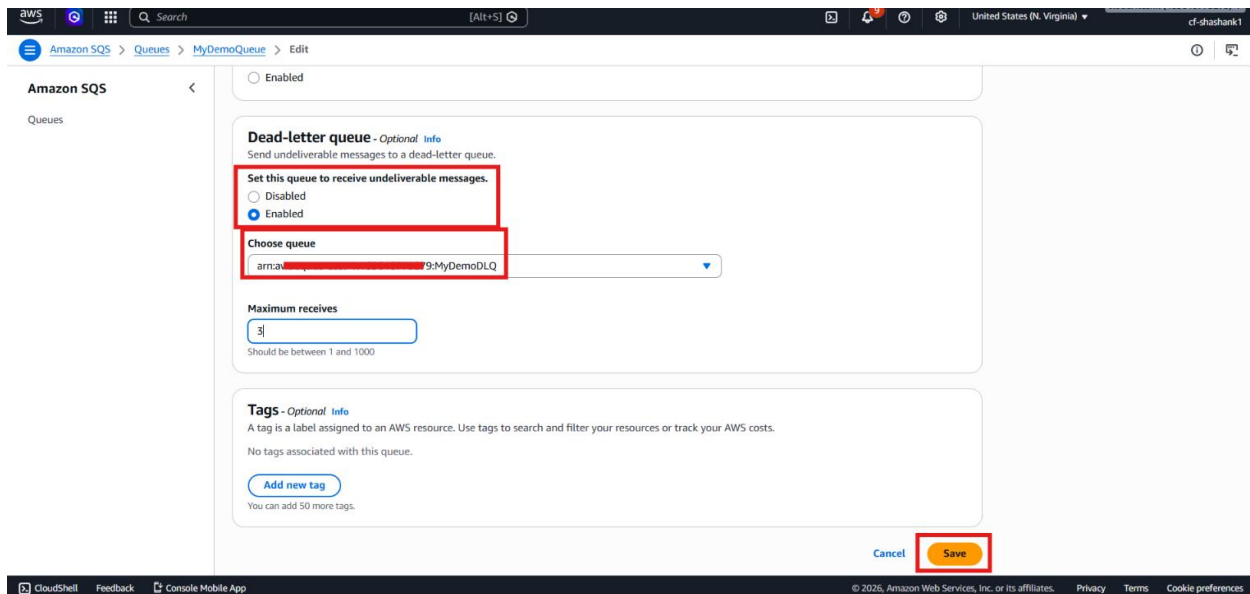
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Access policy [Info](#) [Edit](#)

```
{
  "Version": "2012-10-17",
  "Id": "__default_policy_10",
  "Statement": [
    {
      "Sid": "__owner_statement",
      "Effect": "Allow",
      "Principal": "*",
      "Action": "sqs:*",
      "Resource": "*"
    }
  ]
}
```

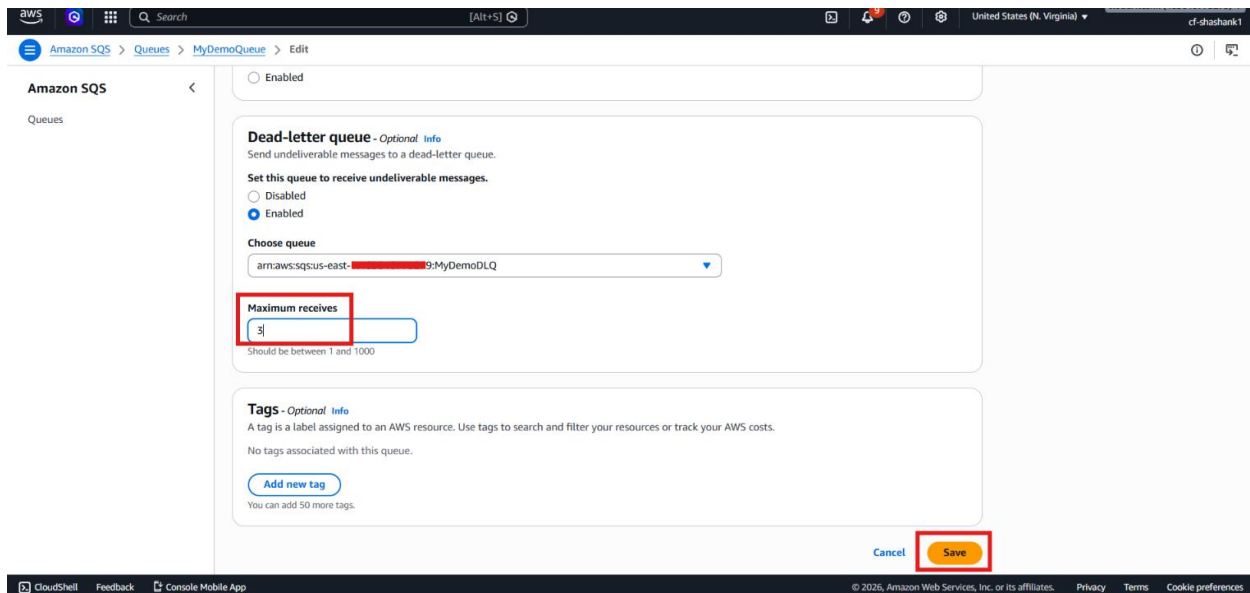
- Open: Dead Letter Queue
- Enable: Dead Letter Queue
- Choose the newly created queue: my-demo-DLQ



- Save the configuration.

Step 4 – Configure Maximum Receive Count

- Edit the DLQ settings.
- Set: Maximum Receives = 3



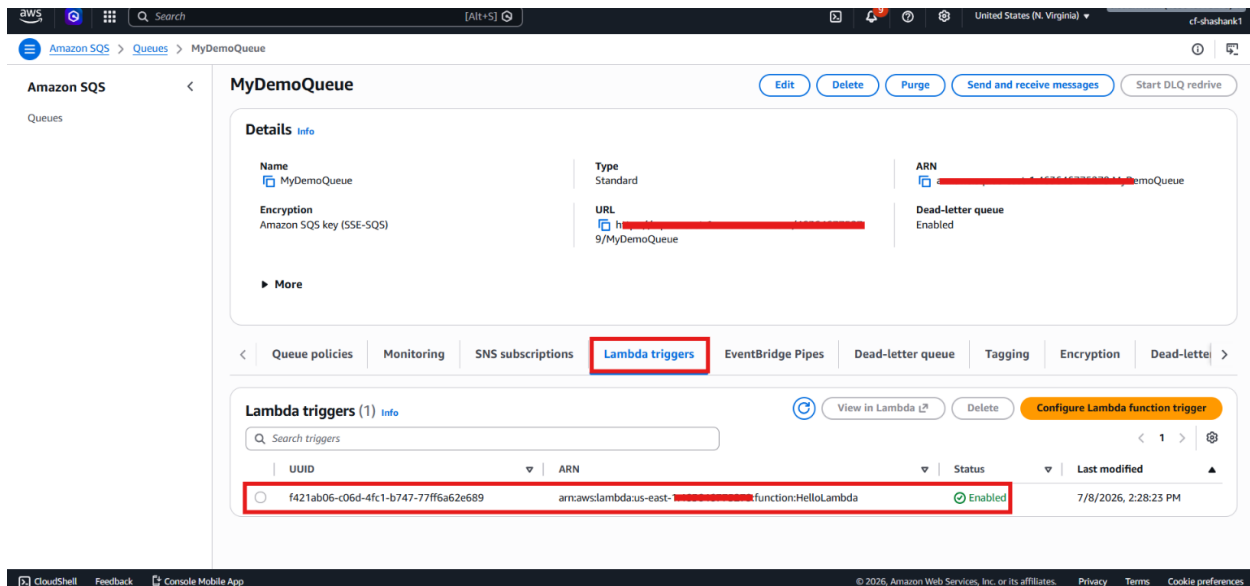
This means:

- Lambda can fail processing the message up to **3 times**.
- After the third failure, SQS automatically moves the message to the DLQ.

AWS allows values between: 1 to 100

Step 5 – Open the Connected Lambda Function

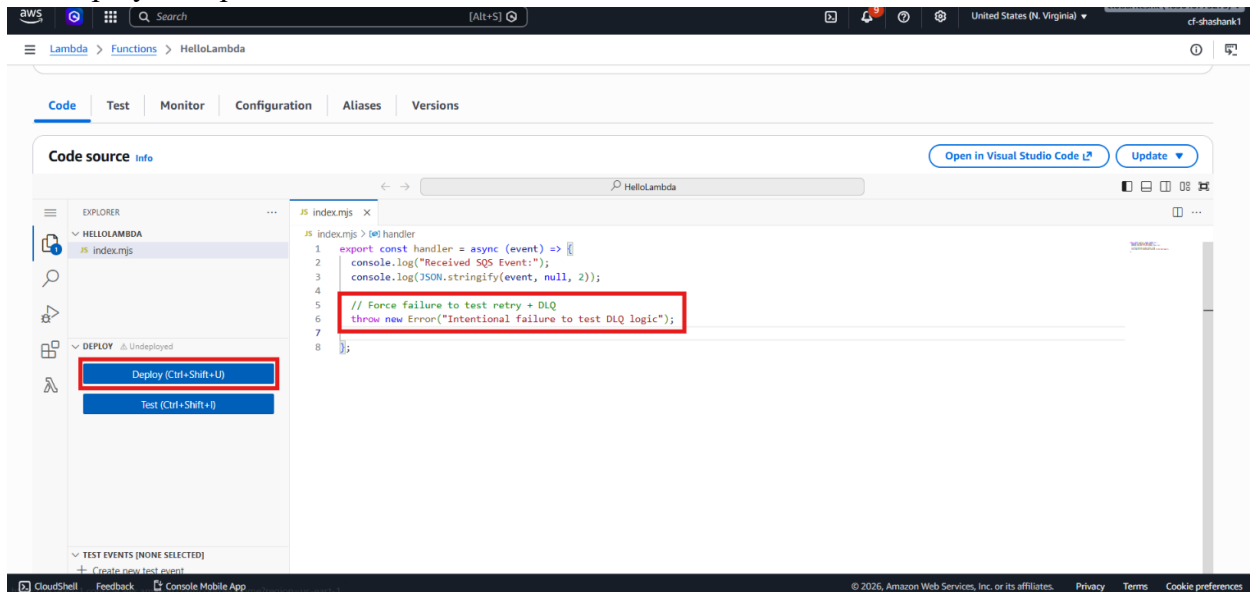
- The main SQS queue is already configured as a trigger for a Lambda function.



- Open the Lambda function connected to the queue.
- Normally, Lambda processes the message successfully.
- For testing, intentionally add an exception inside the Lambda code:

```
// Force failure to test retry + DLQ
throw new Error("Intentional failure to test DLQ logic");
```

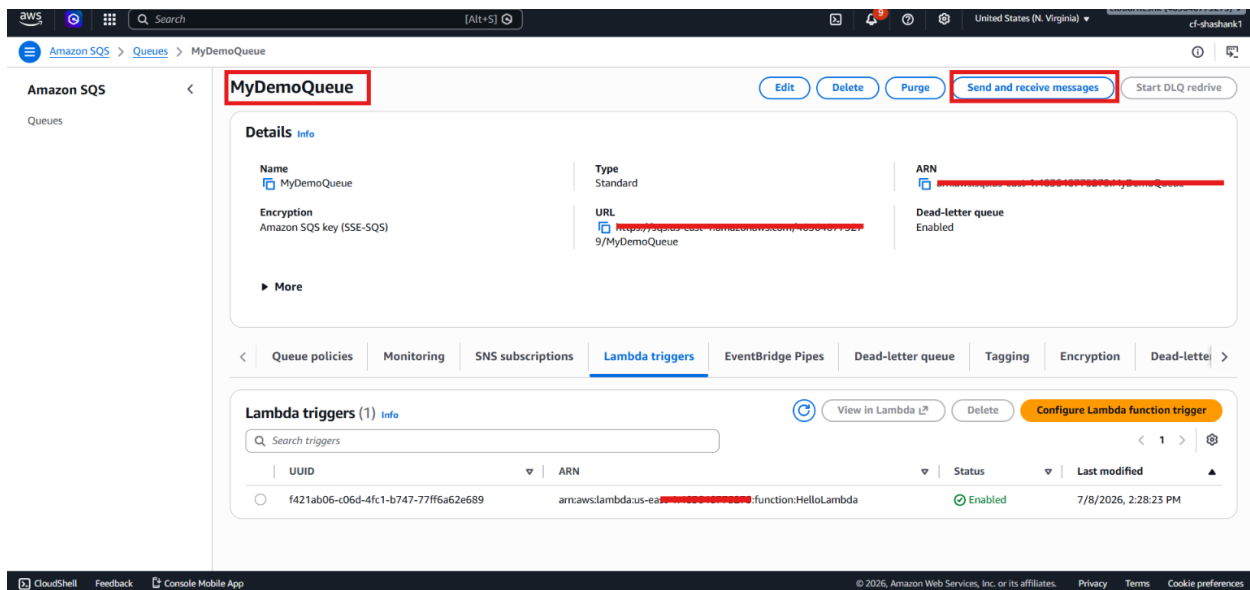
- Deploy the updated Lambda function.



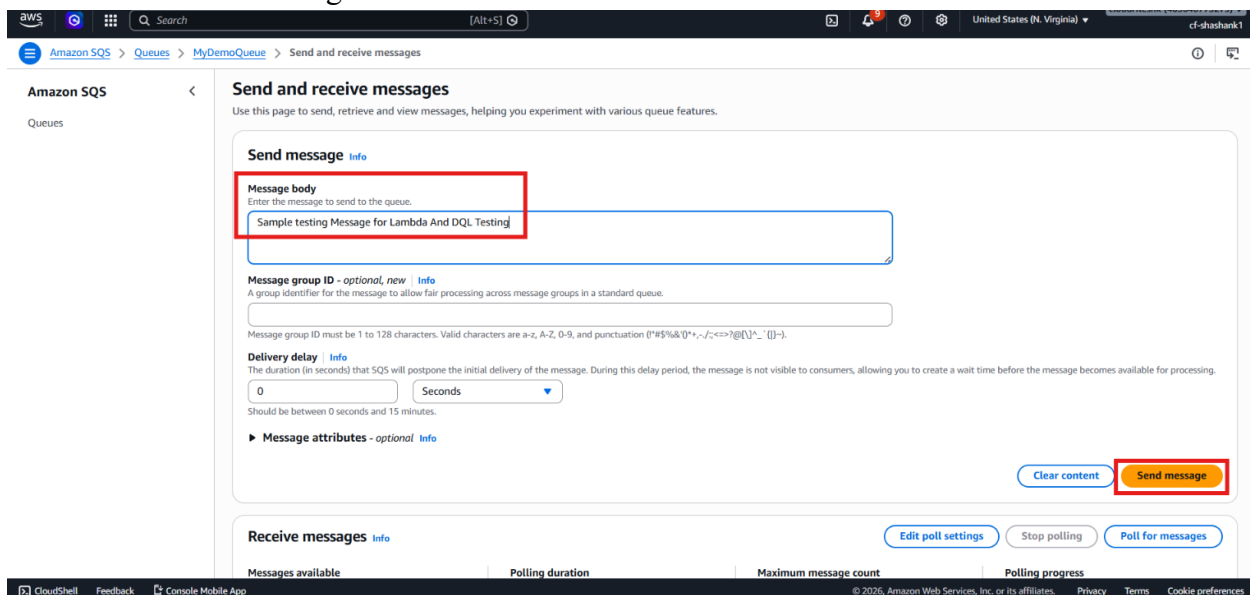
- This simulates a processing failure.

Step 6 – Send a Message to the Main Queue

- Open the **Main Queue**.
- Select: Send Message



- Enter any sample message.
- Click on Send Message.



- The message is now delivered to Lambda through the SQS trigger.

Step 7 – Observe Lambda Failure

Since the Lambda function now throws an exception:

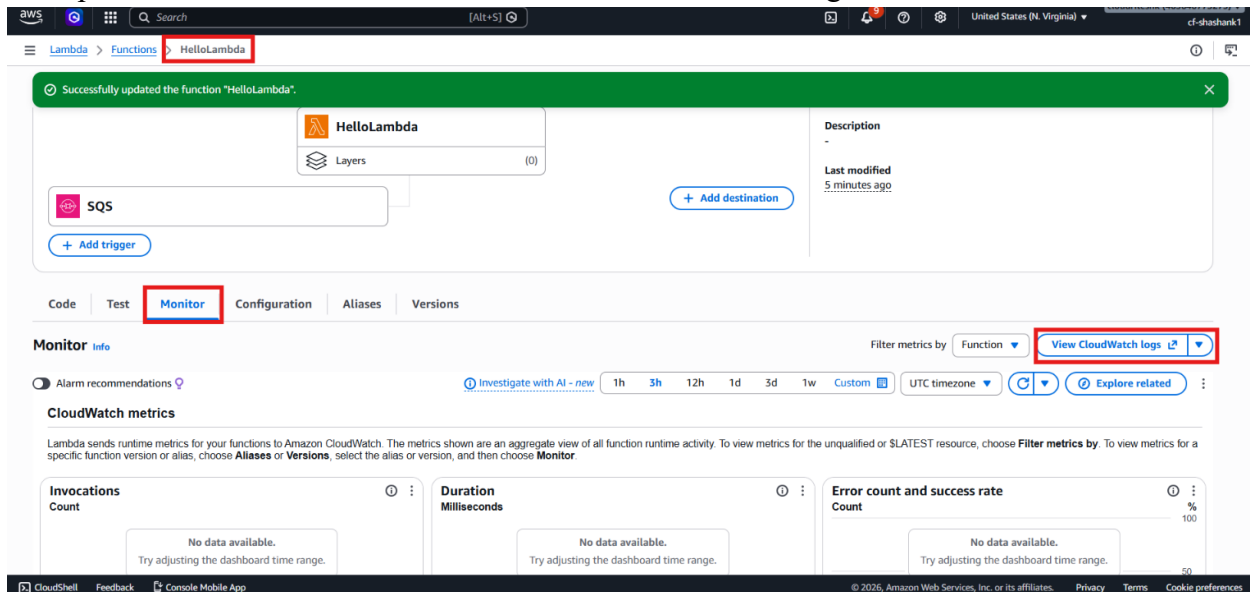
- Lambda receives the message.
- Logs the incoming event.
- Throws an error.
- Processing fails.

SQS automatically retries the message.

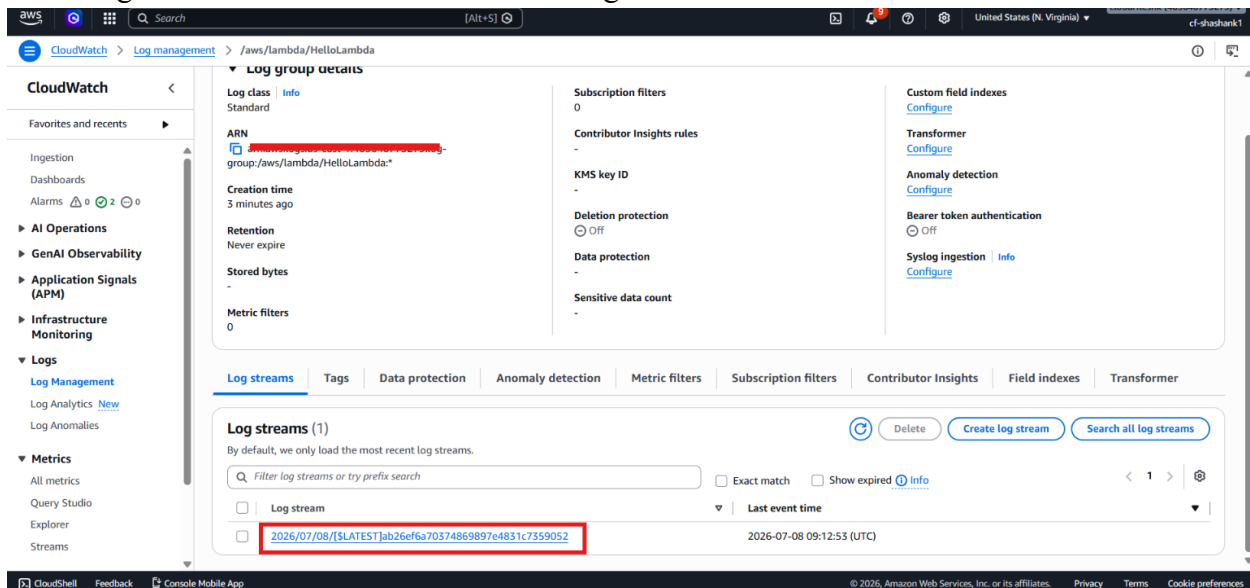
Because the **Maximum Receive Count** is 3, Lambda attempts to process the same message three times before giving up.

Step 8 – Verify CloudWatch Logs

- Open Lambda Function → Monitor → View CloudWatch logs.

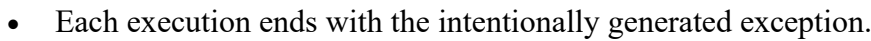


- Navigate to the latest Lambda execution log.



You should notice:

- The same message appears **three times**.



- Open the DLQ.
- Navigate to Send and Receive Messages

Amazon SQS > Queues > MyDemoDLQ

Amazon SQS Queues

MyDemoDLQ

Edit Delete Purge **Send and receive messages** Start DLQ redrive

Details Info

Name	MyDemoDLQ	Type	Standard	ARN	arn:aws:sqs:us-east-1:123456789012:MyDemoDLQ
Encryption	Amazon SQS key (SSE-SQS)	URL	https://sqs.us-east-1.amazonaws.com/123456789012/MyDemoDLQ	Dead-letter queue	-

► More

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Lambda triggers (0) Info

Search triggers

UID ARN Status Last modified

No AWS Lambda triggers

Set incoming messages to trigger a Lambda function.

Configure Lambda function trigger

- Click Poll for Messages

Amazon SQS > Queues > MyDemoDLQ > Send and receive messages

Amazon SQS Queues

Message group ID must be 1 to 128 characters. Valid characters are a-z, A-Z, 0-9, and punctuation (!"#%&'()*+,-./:;<=>?@[\]^_`{|}~).

Delivery delay Info

The duration (in seconds) that SQS will postpone the initial delivery of the message. During this delay period, the message is not visible to consumers, allowing you to create a wait time before the message becomes available for processing.

0 Seconds

Should be between 0 seconds and 15 minutes.

► Message attributes - optional Info

Clear content Send message

Receive messages Info

Messages available: 1 Polling duration: 30 Maximum message count: 10 Polling progress: 0 receives/second 0%

Messages (0)

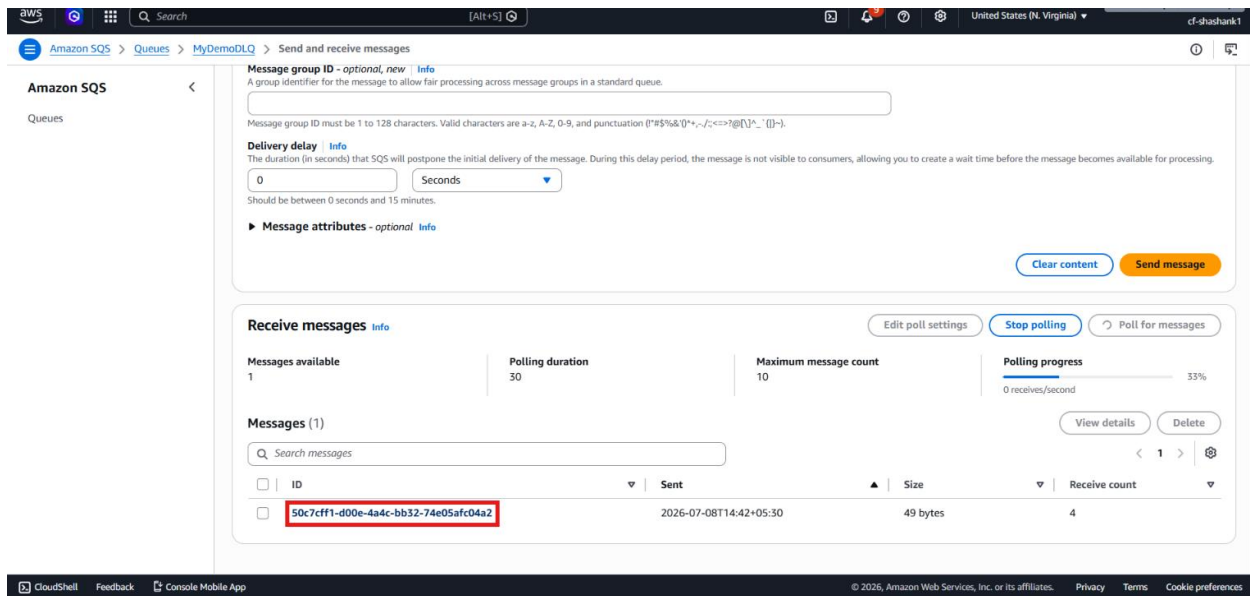
Search messages

ID Sent Size Receive count

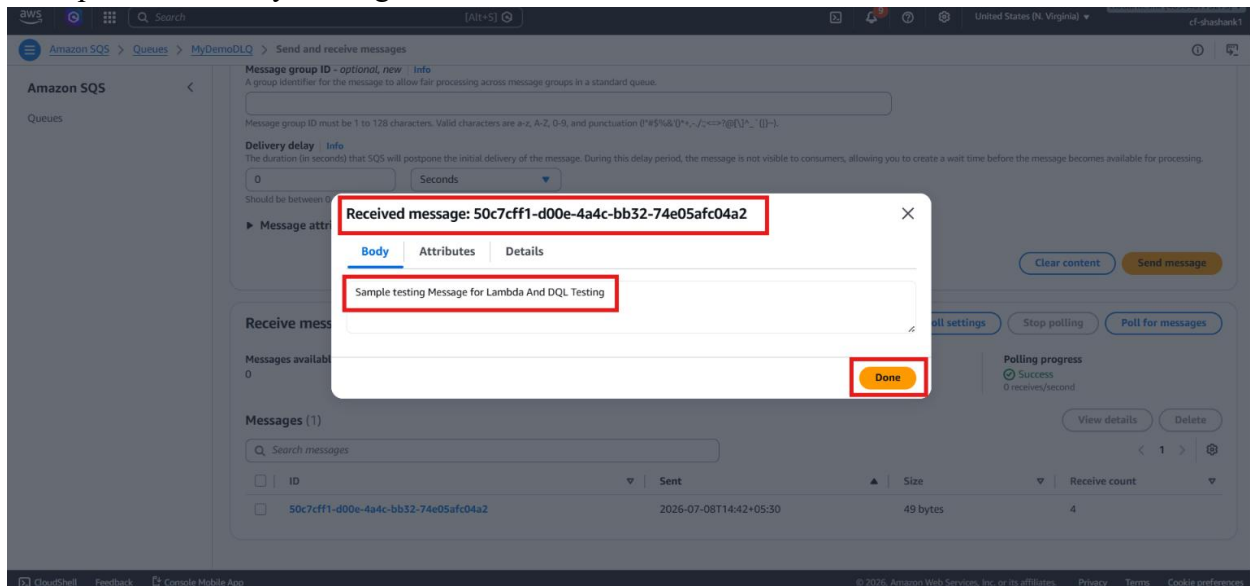
No messages. To view messages in the queue, poll for messages.

Poll for messages

- The failed message should now appear inside the Dead Letter Queue.



- Open it and verify message.



- This confirms that after three failed processing attempts, SQS automatically transferred the message to the DLQ.

Verification

- Main SQS queue is configured with a Dead Letter Queue.
- Maximum Receive Count is set (3 in this lab).
- Lambda throws an intentional exception.
- The same message is retried three times.
- CloudWatch Logs show three processing attempts.
- The failed message appears inside the Dead Letter Queue.
- Polling the DLQ successfully retrieves the failed message.